

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

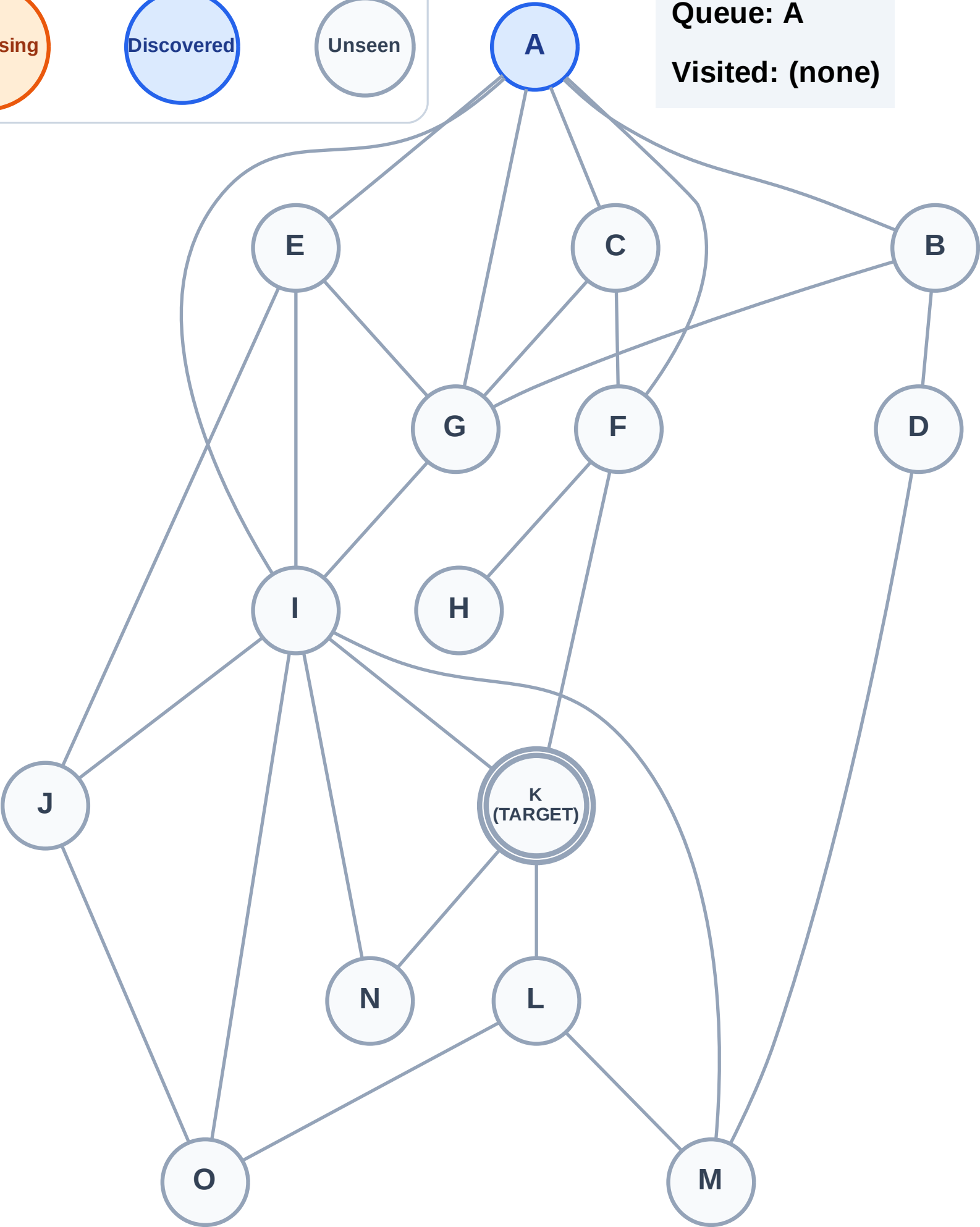
Processing

Discovered

Unseen

Queue: A

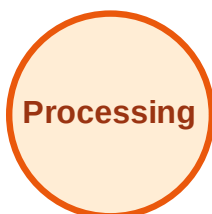
Visited: (none)



BFS Traversal

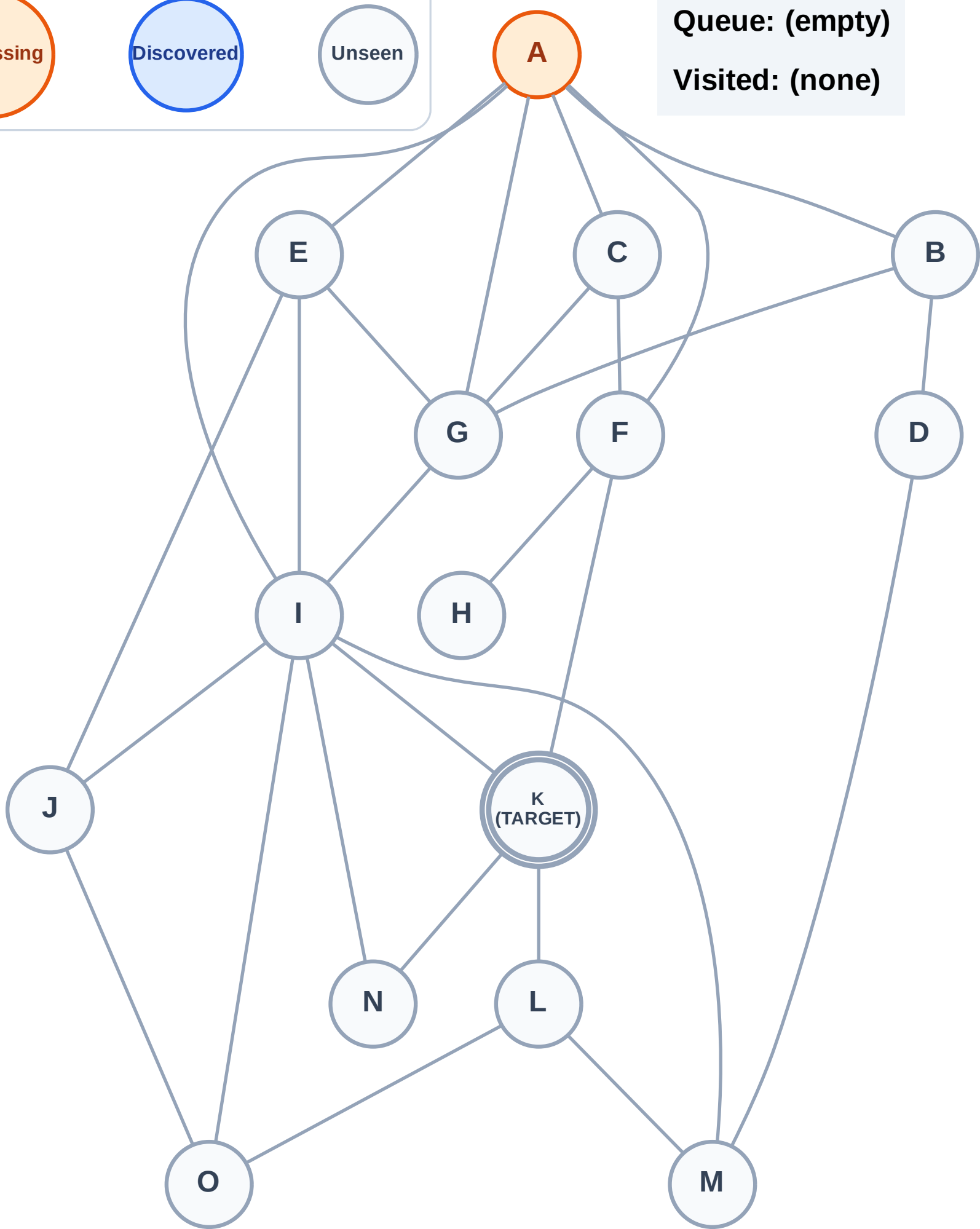
Step 2: Process node A

Legend



Queue: (empty)

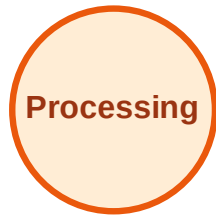
Visited: (none)



BFS Traversal

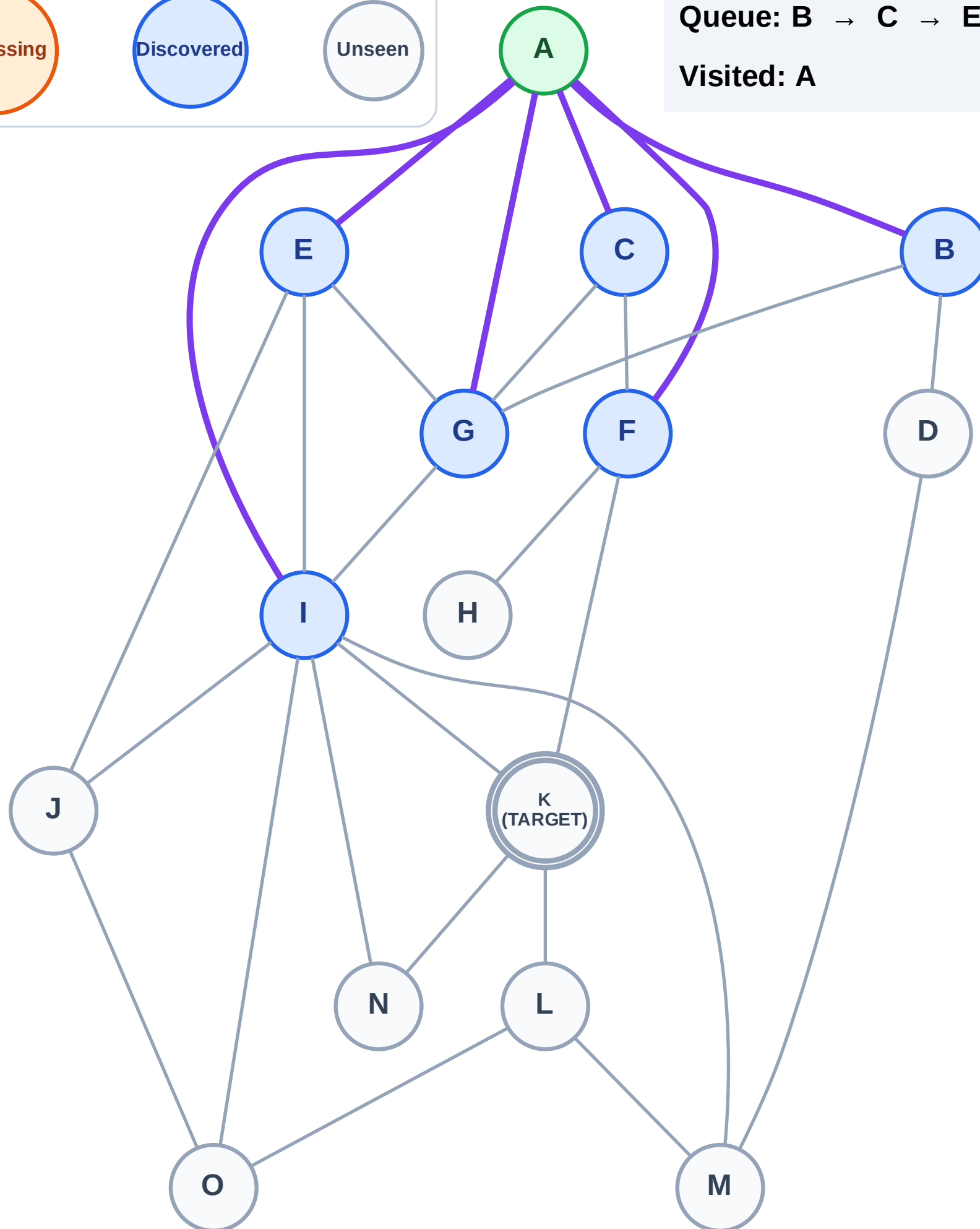
Step 3: Discover B, C, E, F, G, I from A

Legend



Queue: B → C → E → F → G → I

Visited: A



BFS Traversal

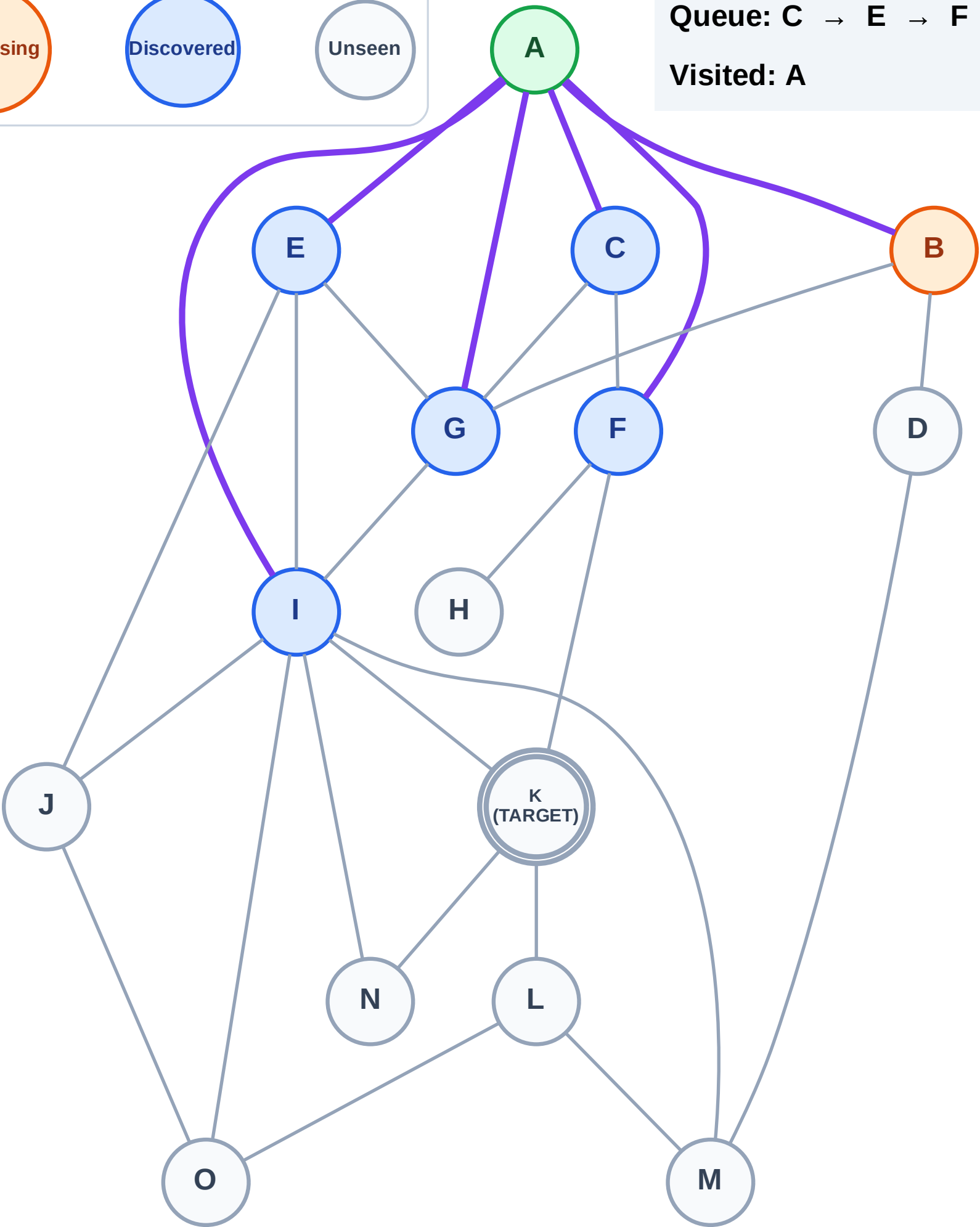
Step 4: Process node B

Legend



Queue: C → E → F → G → I

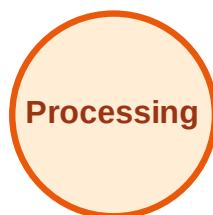
Visited: A



BFS Traversal

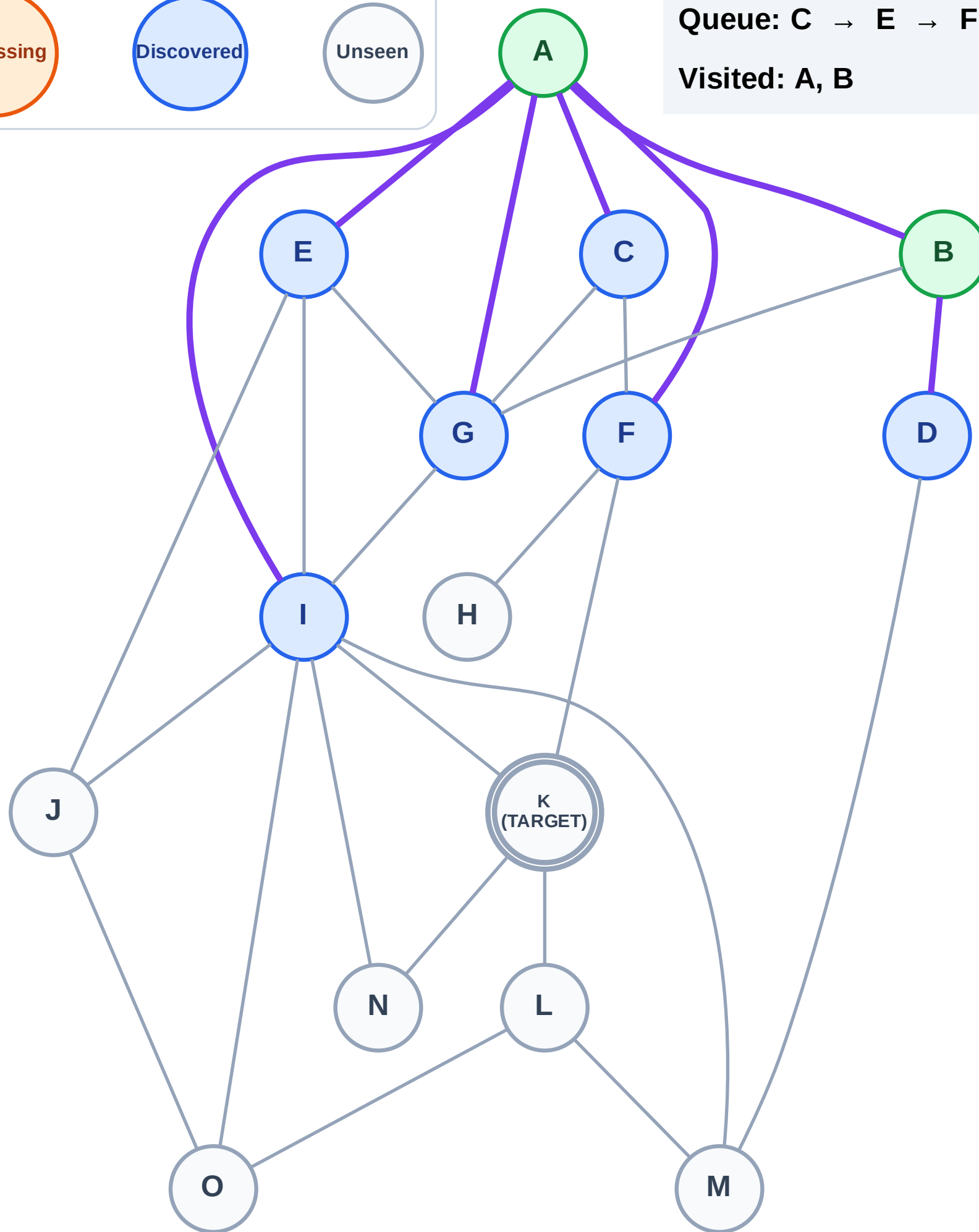
Step 5: Discover D from B

Legend



Queue: C → E → F → G → I → D

Visited: A, B



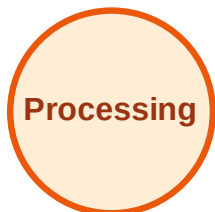
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



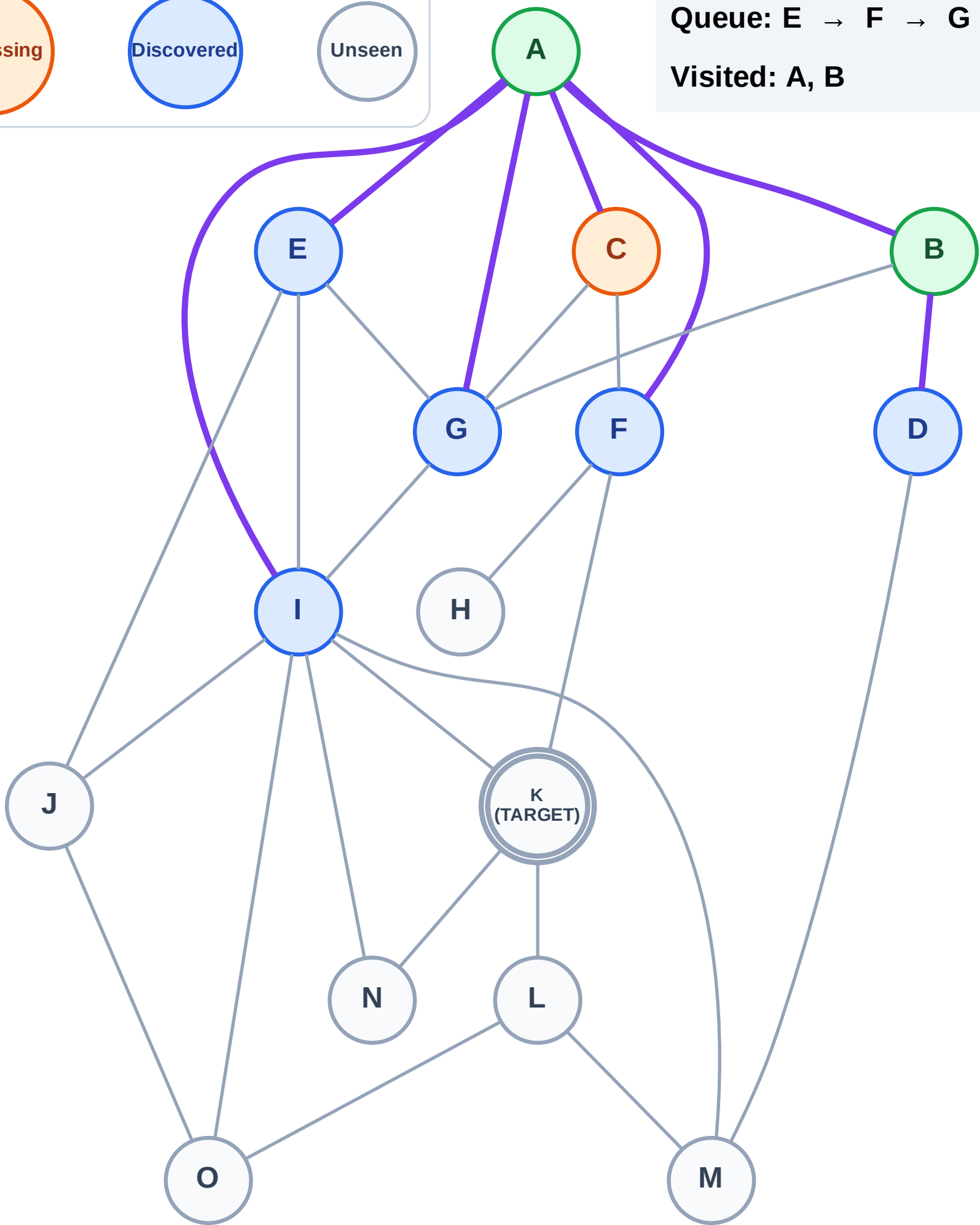
Discovered



Unseen

Queue: E → F → G → I → D

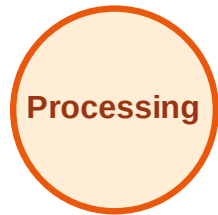
Visited: A, B



BFS Traversal

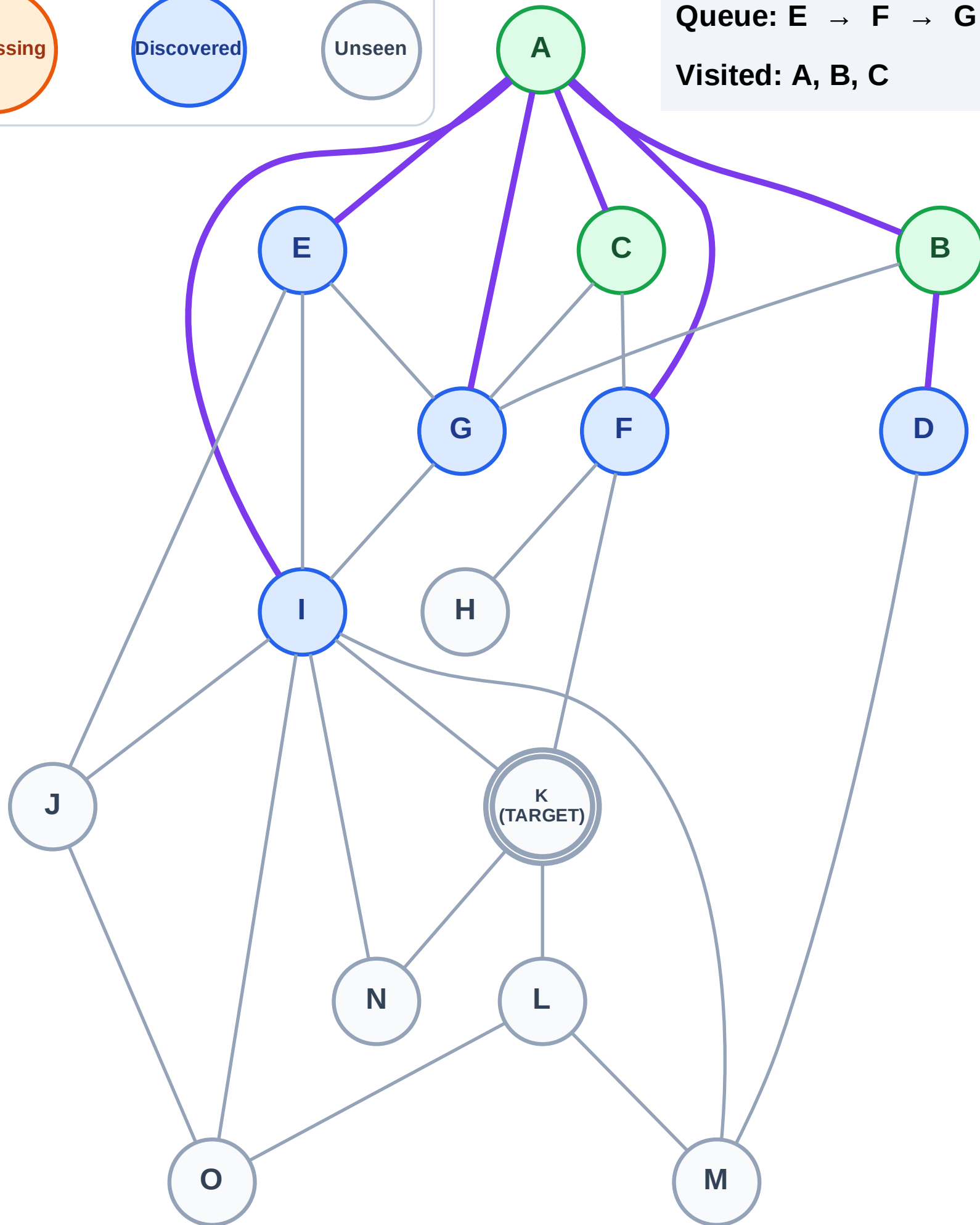
Step 7: No new neighbors from C

Legend



Queue: E → F → G → I → D

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

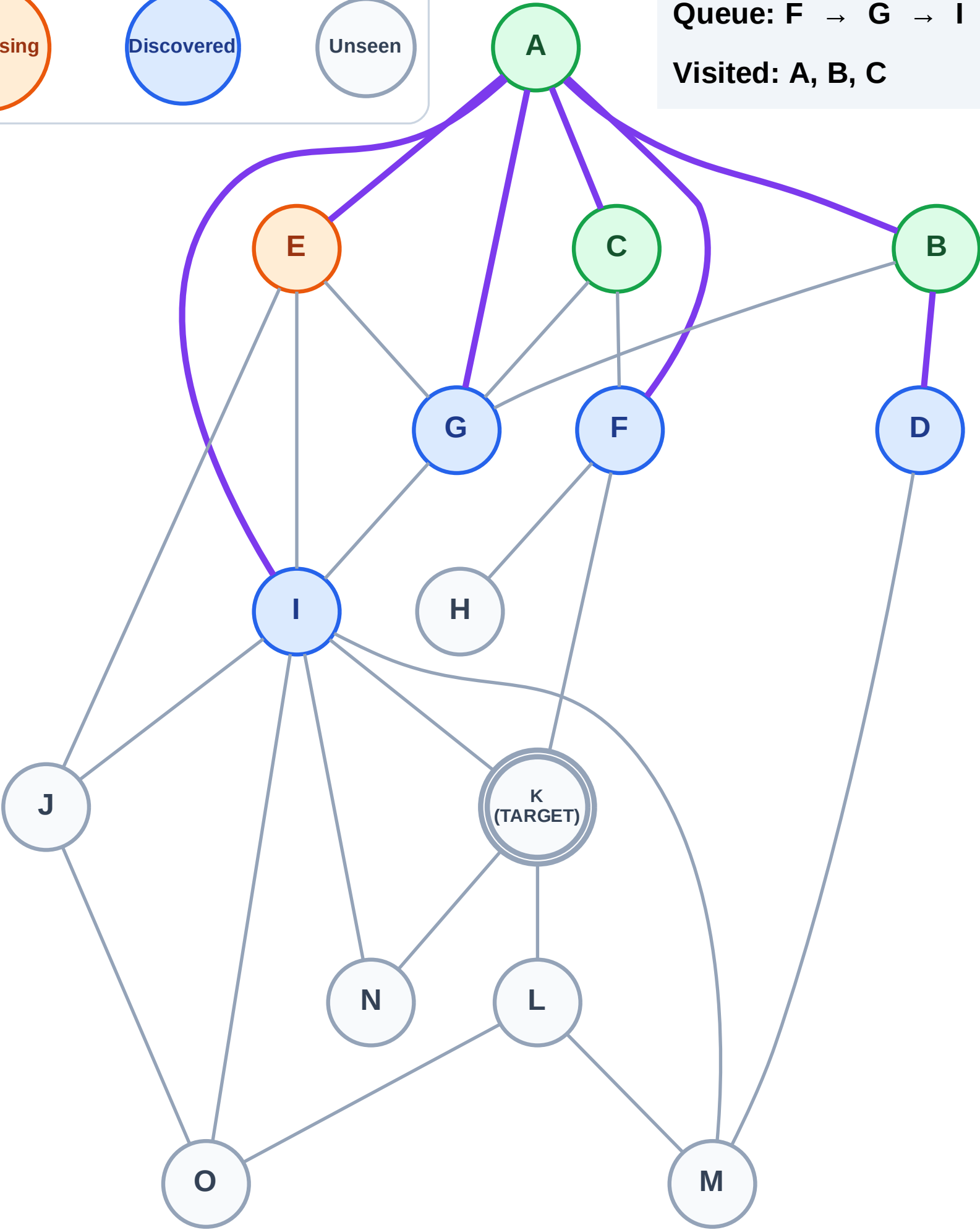
Processing

Discovered

Unseen

Queue: F → G → I → D

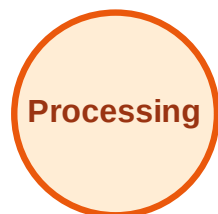
Visited: A, B, C



BFS Traversal

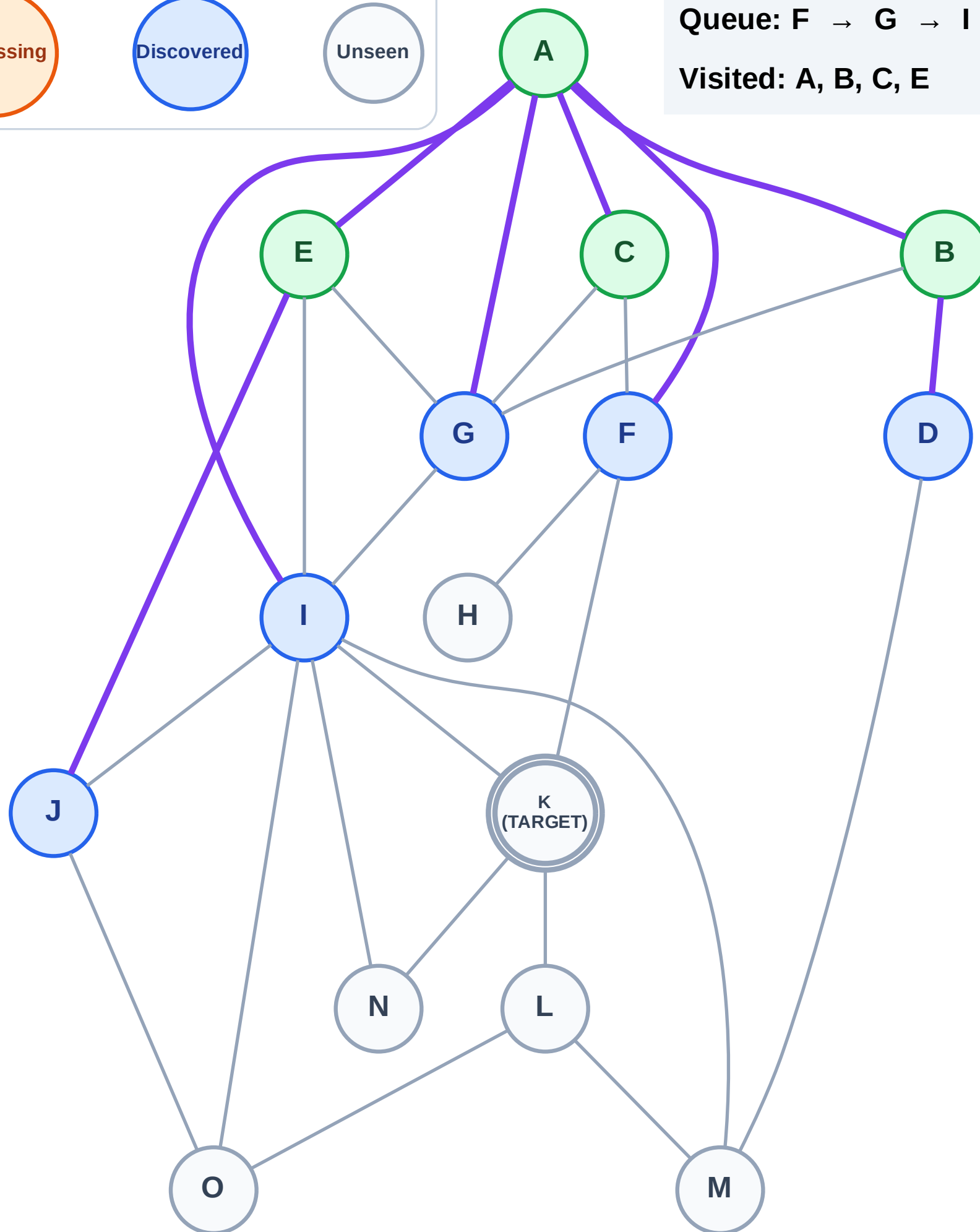
Step 9: Discover J from E

Legend



Queue: F → G → I → D → J

Visited: A, B, C, E



BFS Traversal

Step 10: Process node F

Legend

Complete

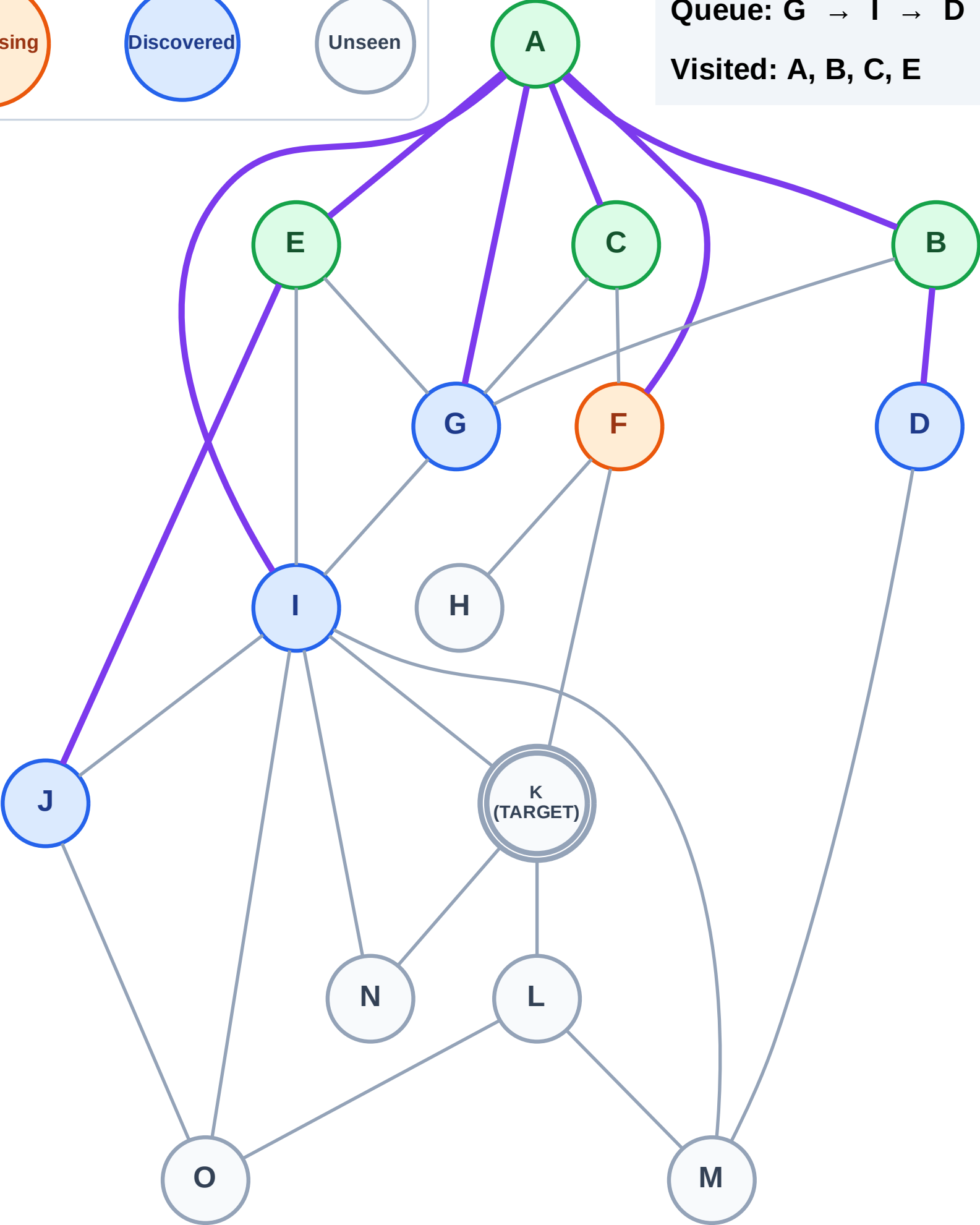
Processing

Discovered

Unseen

Queue: G → I → D → J

Visited: A, B, C, E



BFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → F → K

Legend

Complete

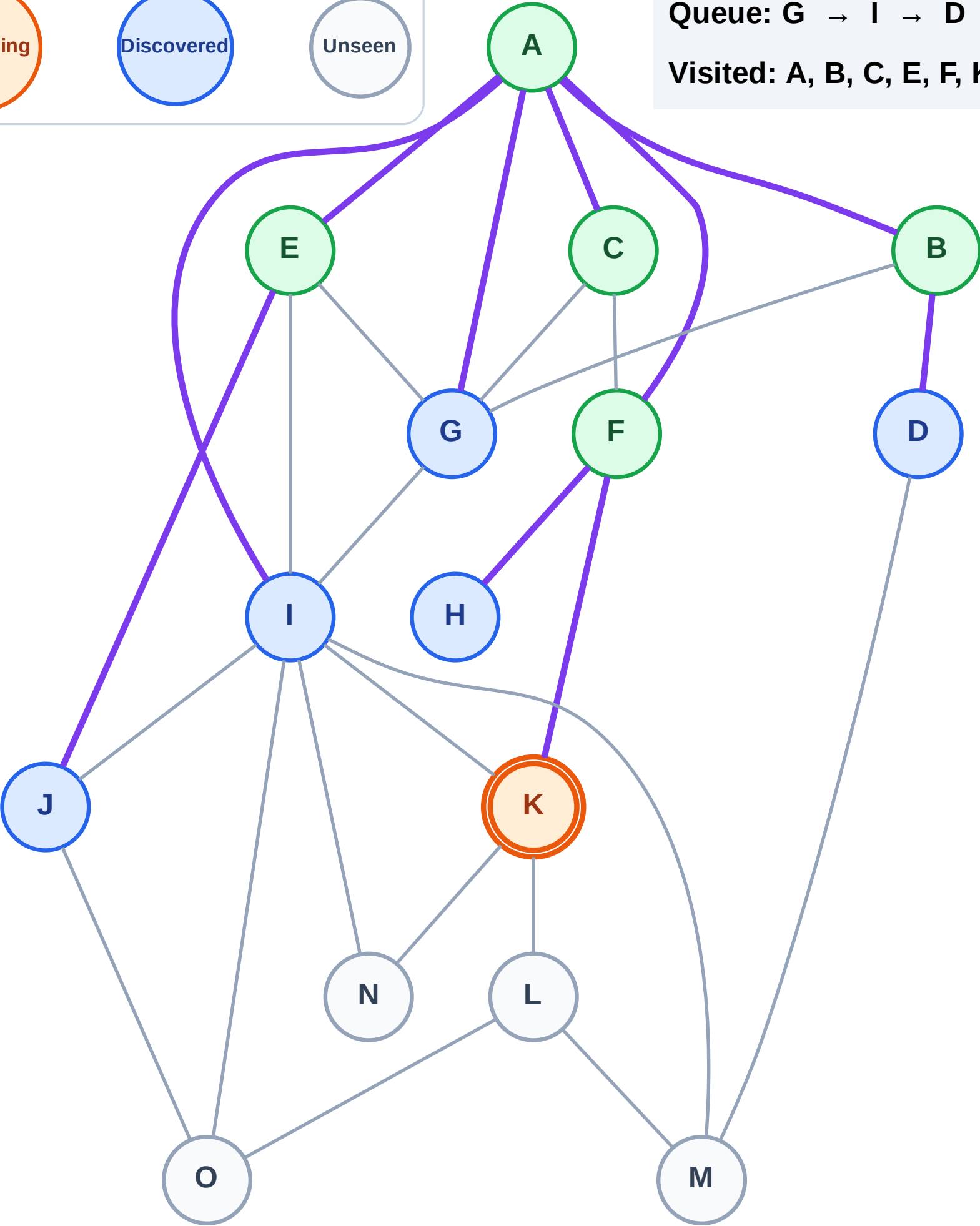
Processing

Discovered

Unseen

Queue: G → I → D → J → H

Visited: A, B, C, E, F, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

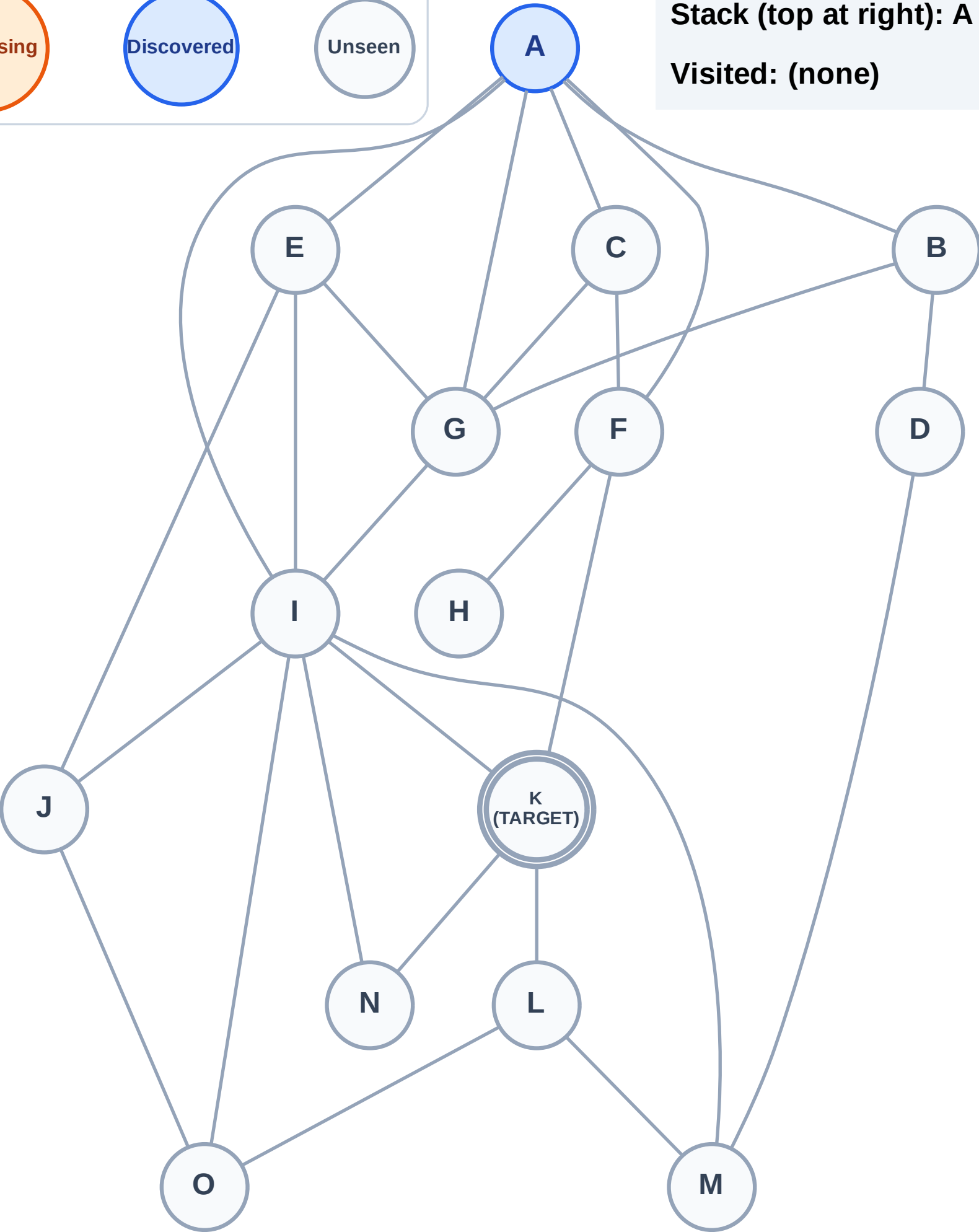
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

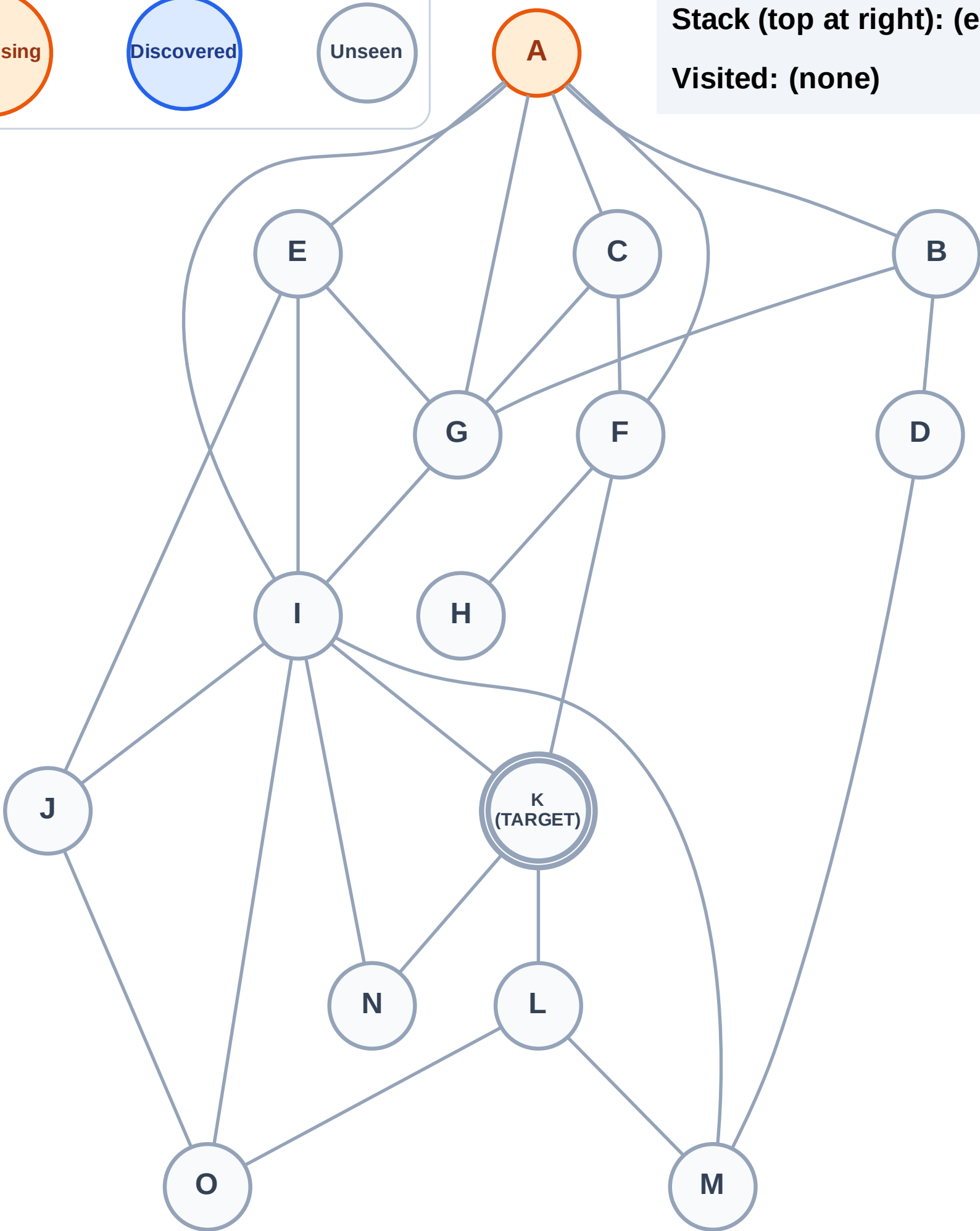
Processing

Discovered

Unseen

Stack (top at right): (empty)

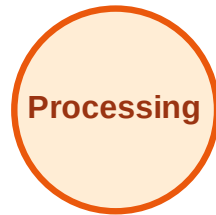
Visited: (none)



DFS Traversal

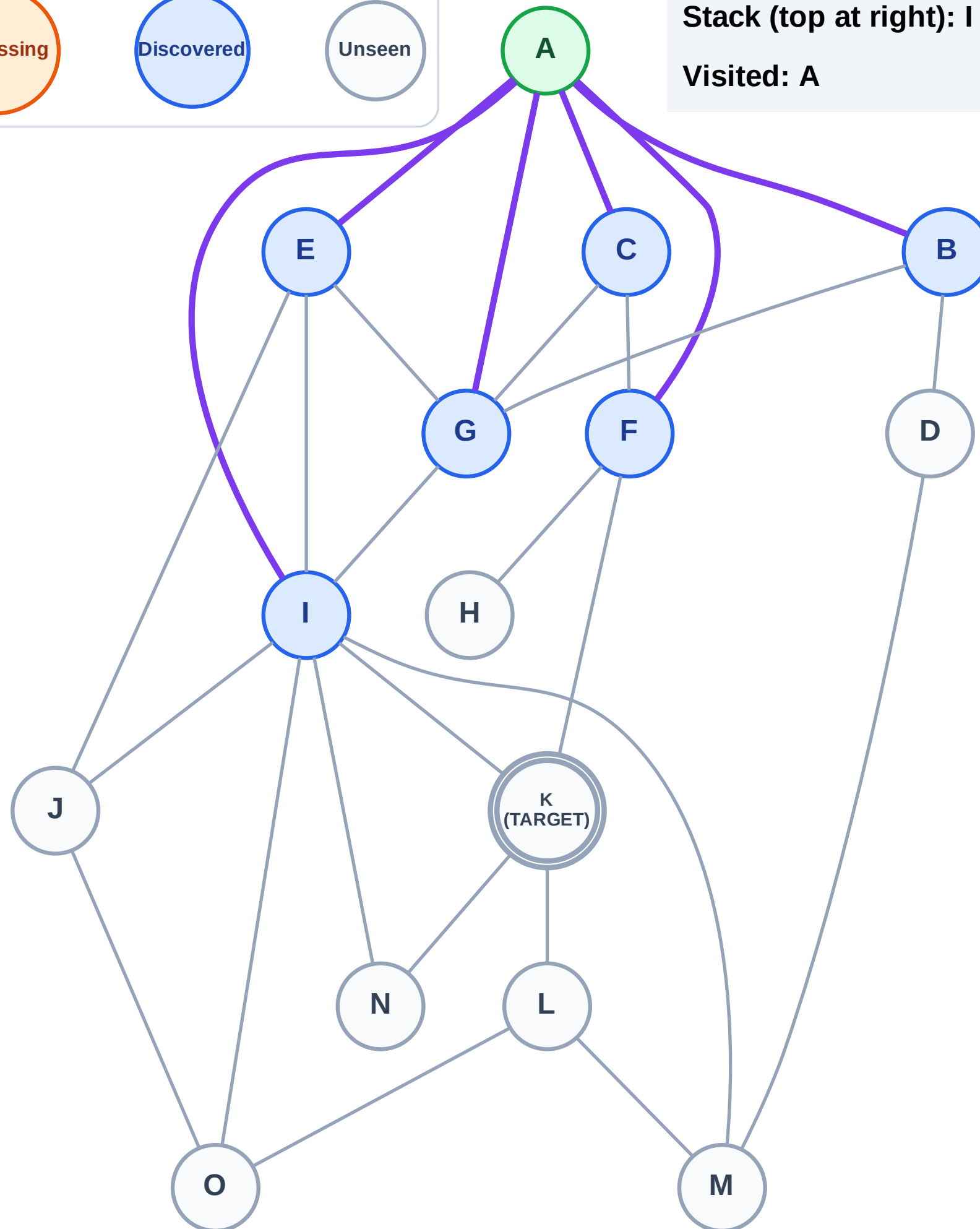
Step 3: Discover I, G, F, E, C, B from A

Legend



Stack (top at right): I | G | F | E | C | B

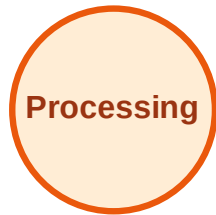
Visited: A



DFS Traversal

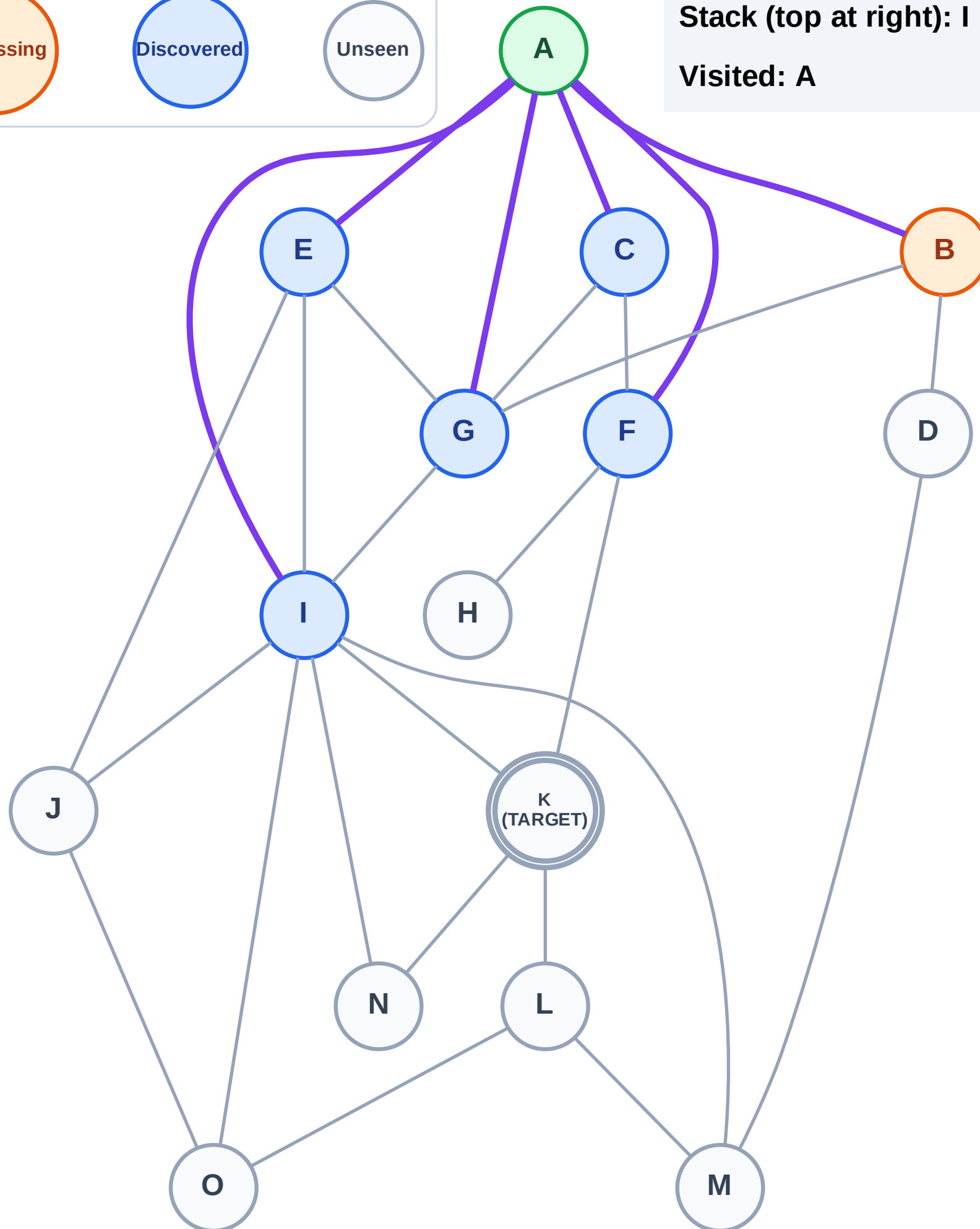
Step 4: Process node B

Legend



Stack (top at right): I | G | F | E | C

Visited: A



DFS Traversal

Step 5: Discover D from B

Legend

Complete

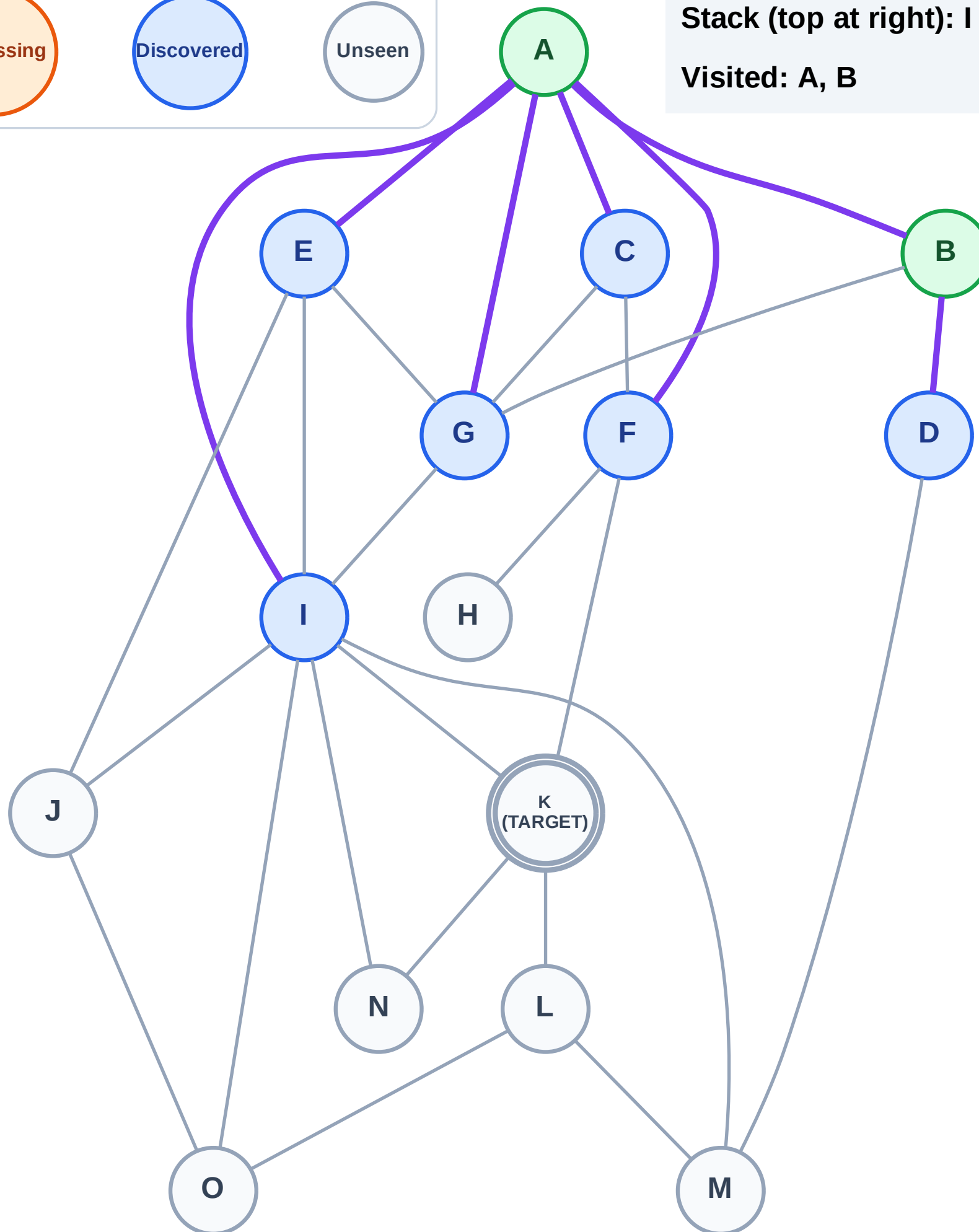
Processing

Discovered

Unseen

Stack (top at right): I | G | F | E | C | D

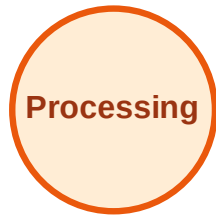
Visited: A, B



DFS Traversal

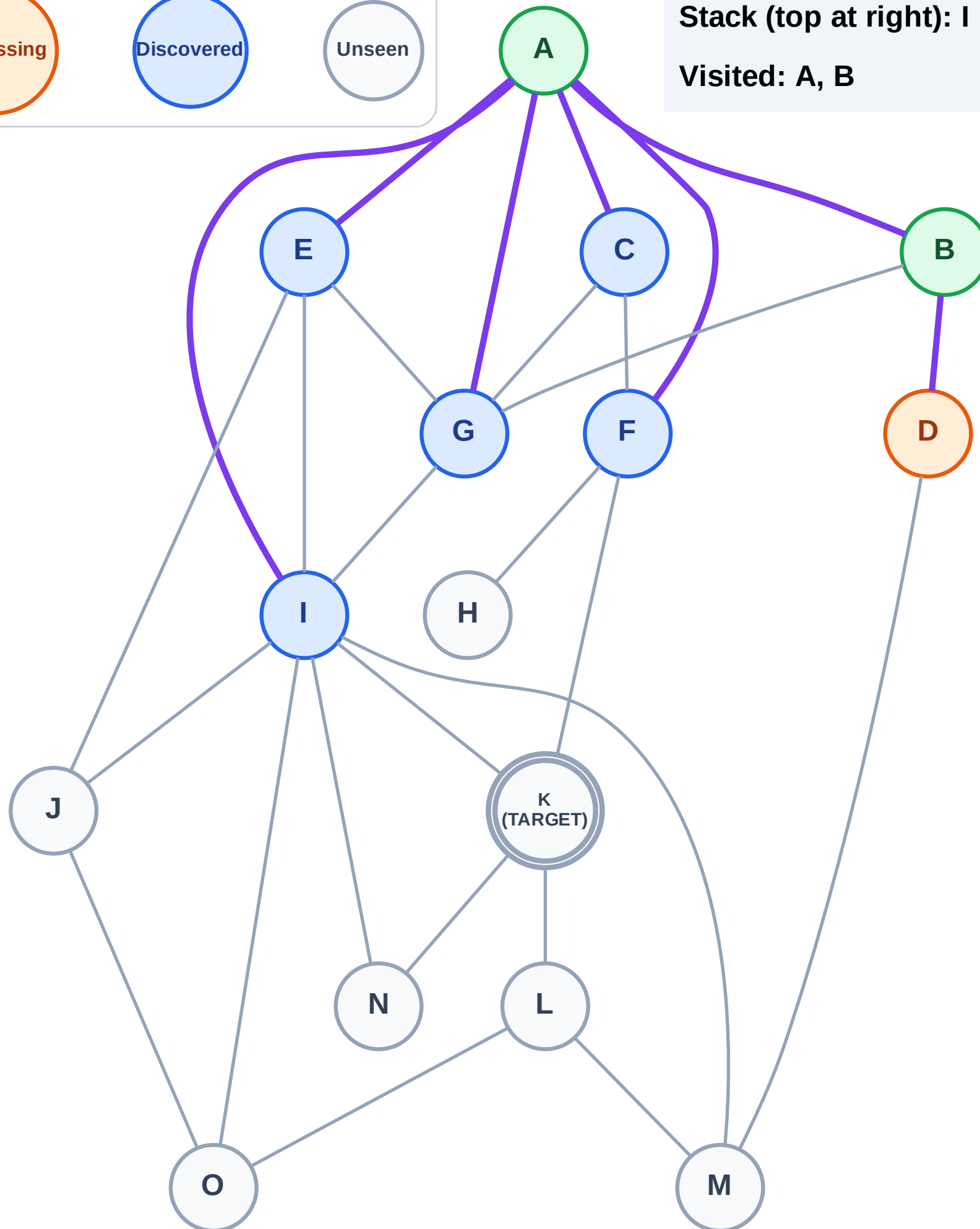
Step 6: Process node D

Legend



Stack (top at right): I | G | F | E | C

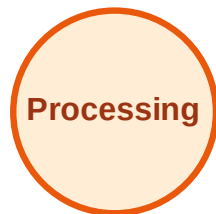
Visited: A, B



DFS Traversal

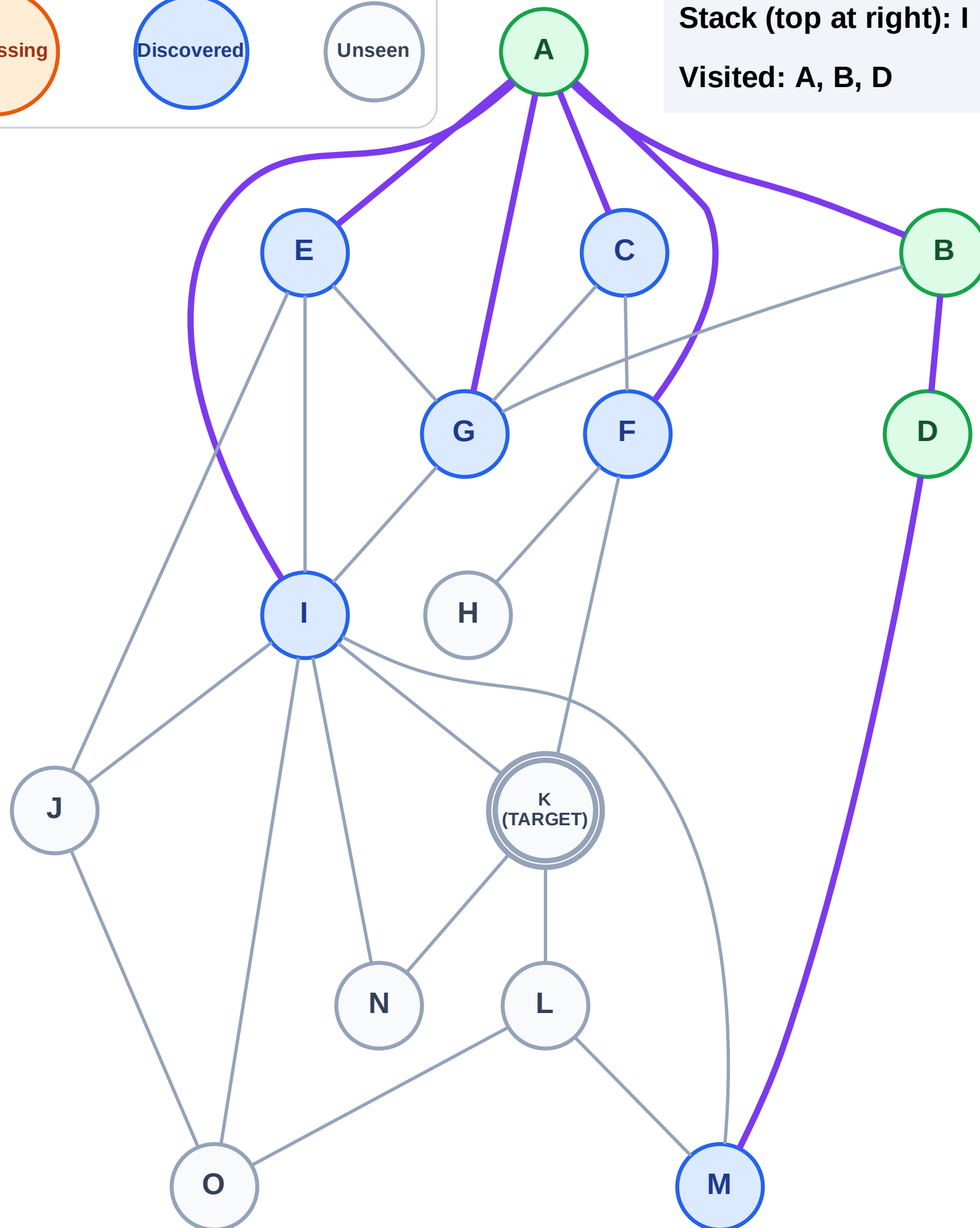
Step 7: Discover M from D

Legend



Stack (top at right): I | G | F | E | C | M

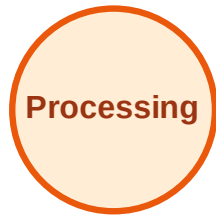
Visited: A, B, D



DFS Traversal

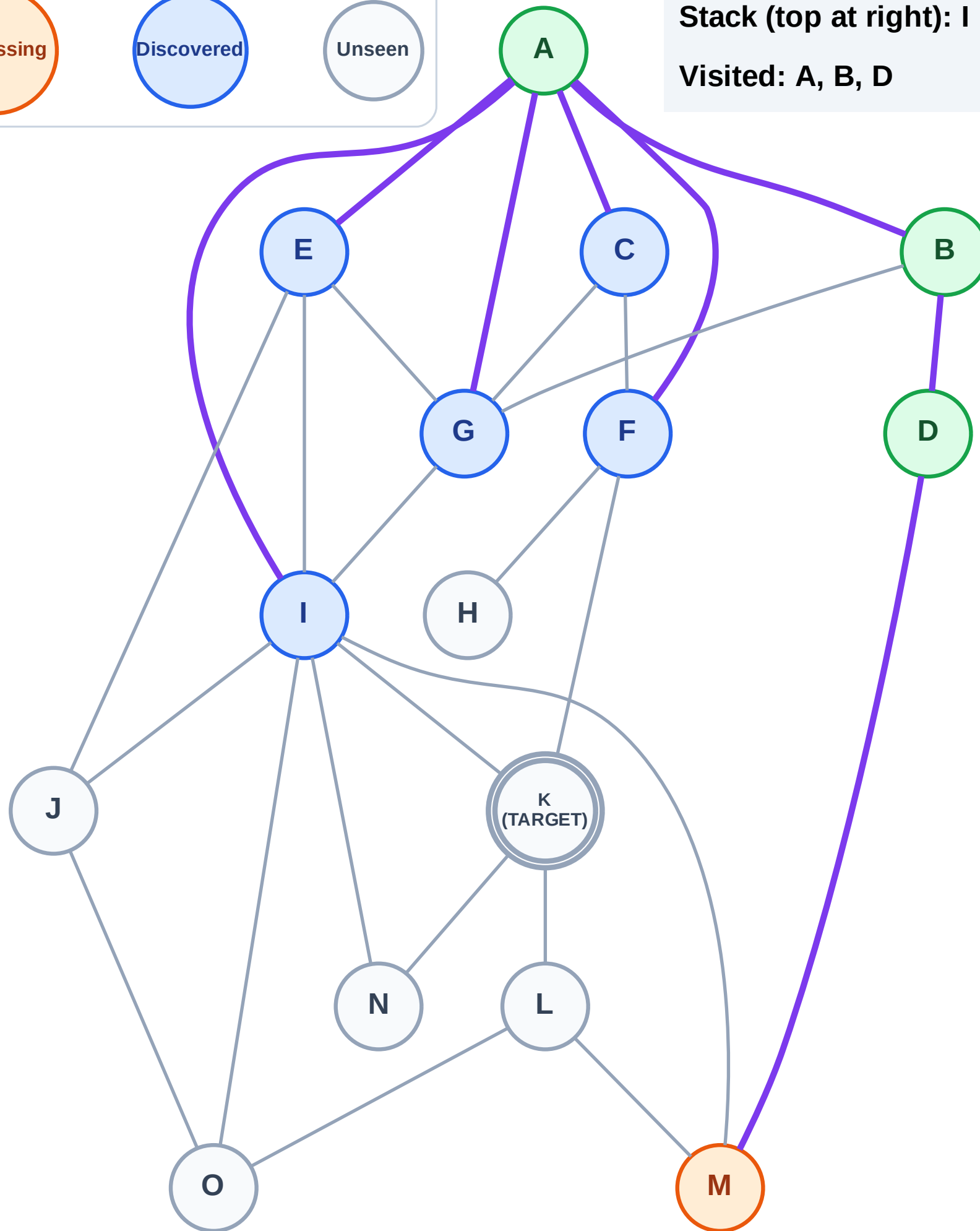
Step 8: Process node M

Legend



Stack (top at right): I | G | F | E | C

Visited: A, B, D



DFS Traversal

Step 9: Discover L from M

Legend

Complete

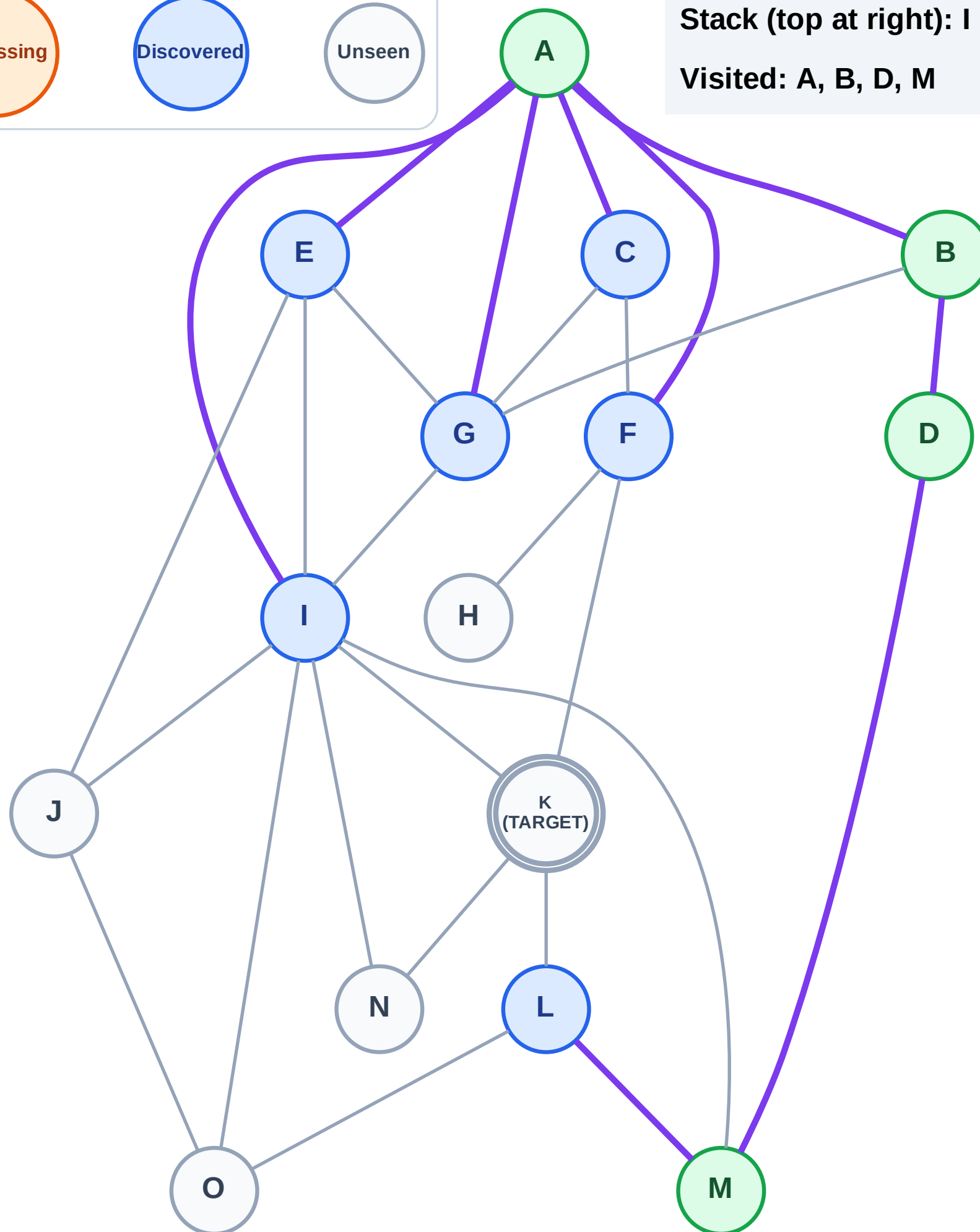
Processing

Discovered

Unseen

Stack (top at right): I | G | F | E | C | L

Visited: A, B, D, M



DFS Traversal

Step 10: Process node L

Legend

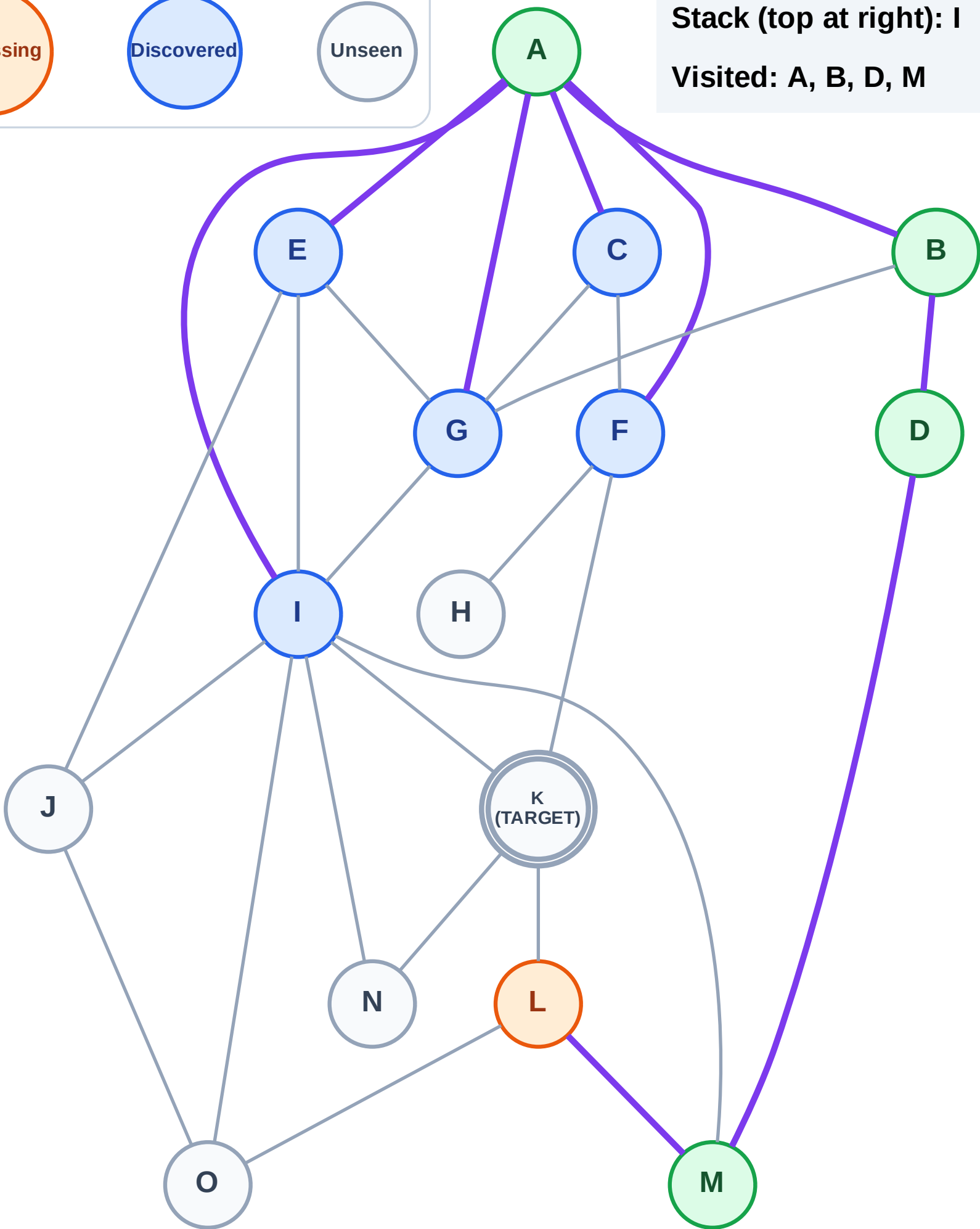
Complete

Processing

Discovered

Unseen

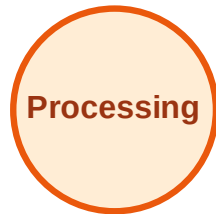
Stack (top at right): I | G | F | E | C
Visited: A, B, D, M



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → D → M → L → K

Legend



Stack (top at right): I | G | F | E | C | O

Visited: A, B, D, K, L, M

